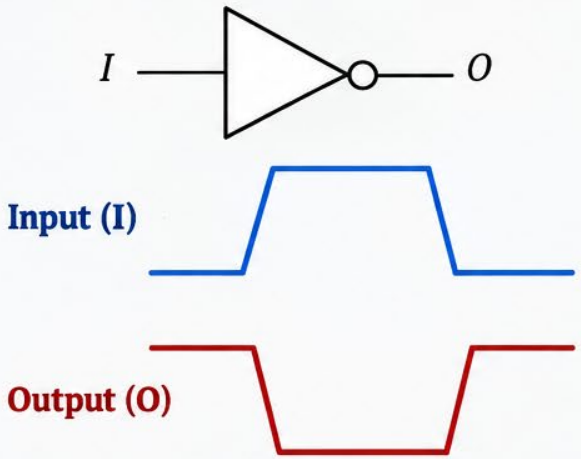
Cell Timing Characterization

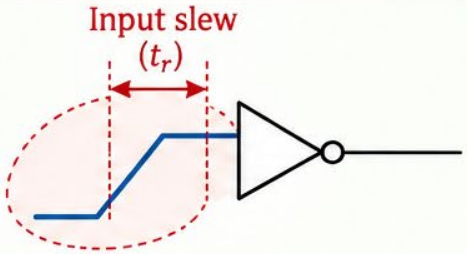
Ranga Vemuri

Effect of Input Slew On Cell Delay

1. Inverter and Input Slew



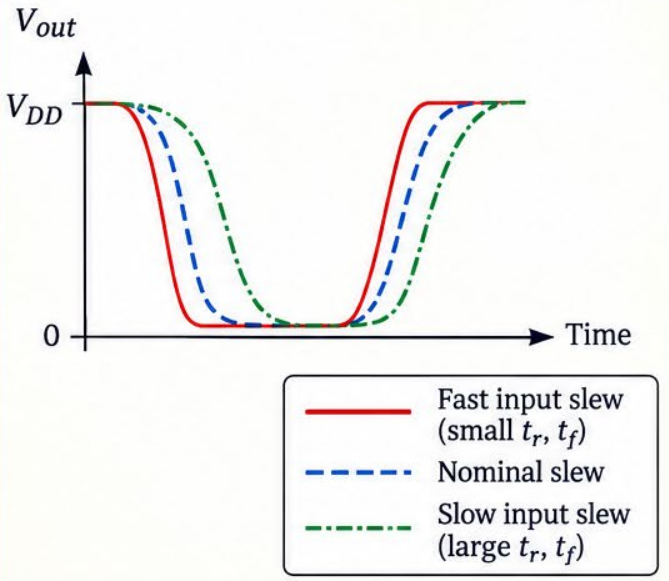
2. Effect of Input Slew



Larger input slew (slower rise/fall) results in a slower output transition and larger cell delay.

Input slew (t_r, t_f):
Time for the input to transition from X% to Y% of V_{DD}.

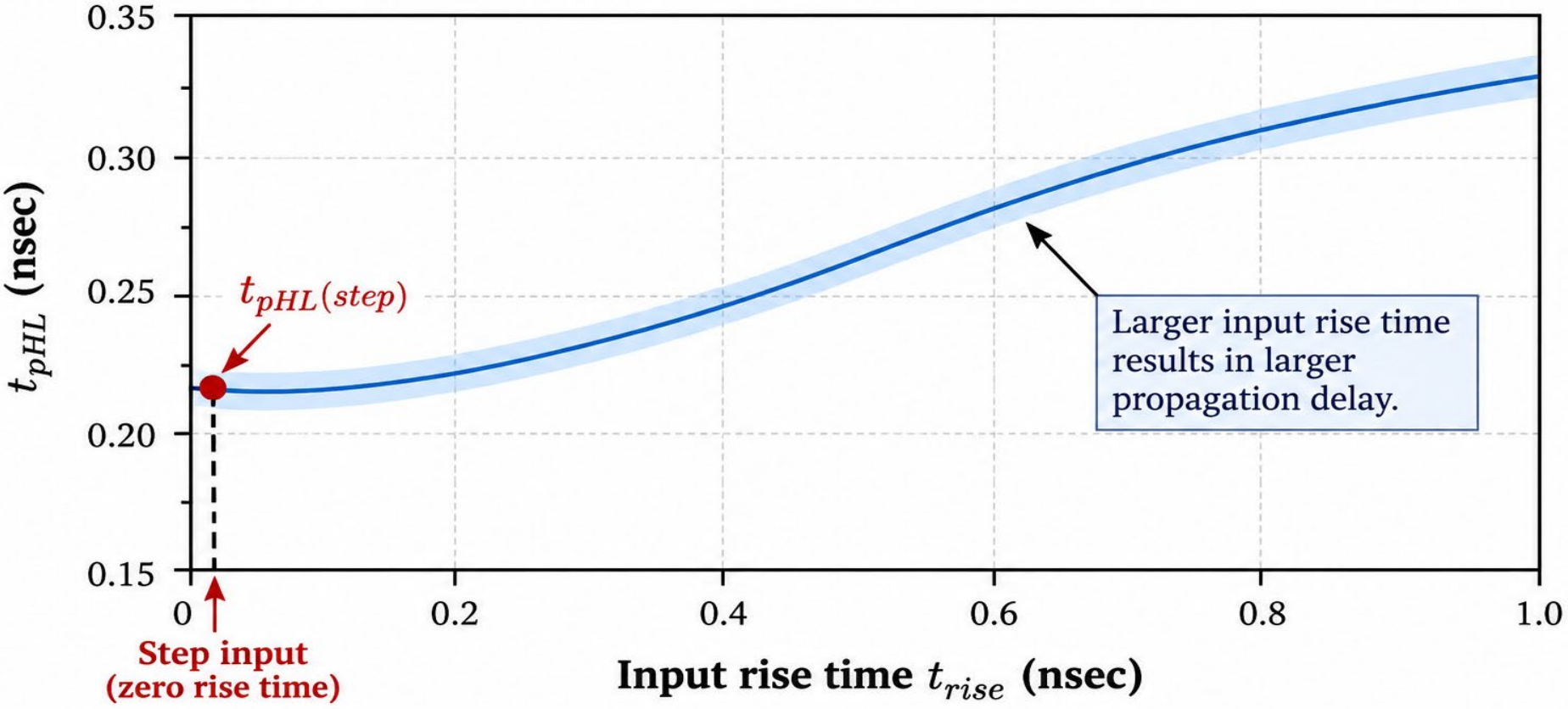
3. Output Waveforms for Different Input Slews



Key Definitions

Slew rate	Rate of change of voltage. Typical value: 100 mV/ps (i.e., 0.1 V/ns).
Slew (transition time)	Time for a signal to transition from X% of V_{DD} to Y% of V_{DD}.
Input slew (or slew)	Rise or fall time at the input of a cell. Typically measured from 10% to 90% of V_{DD}.

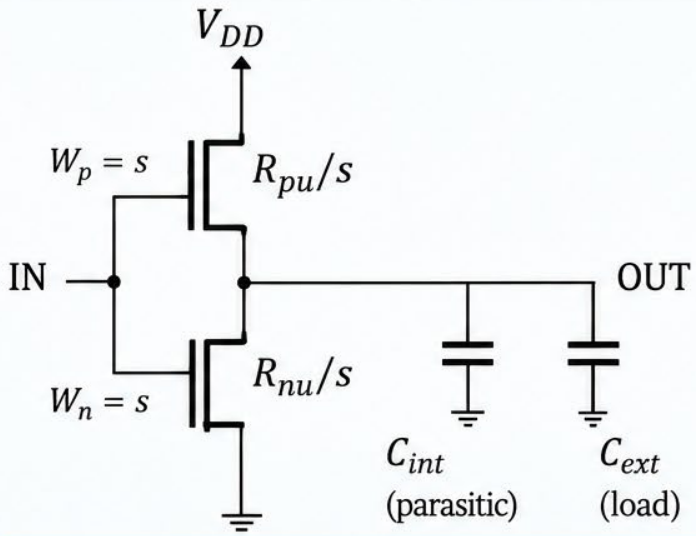
Impact of Rise Time on Delay



Propagation delay model: $$t_{pHL} = \sqrt{t_{pHL(step)}^2 + (t_r/2)^2}$$

Intrinsic Delay vs. Effort Delay of a Unit Inverter
(Due to External Load)

Unit Inverter



s : sizing factor
($W_p = W_n = s$ relative to minimum)

C_{int} includes:
- gate/diffusion capacitances
- contacts and wiring
- metal parasitics
- other parasitics

Unit Transistor Resistances

R_{pu}, R_{nu} : effective resistances of unit PMOS and NMOS transistors

Propagation Delay Components

Intrinsic delay (t_{ri}): $t_{ri} = k_1 R_{pu} C_{int}$

Effort delay (t_{fu}): $t_{fu} = k_2 R_{nu} C_{int}$

Rise and fall delays of a unit inverter (with no external load)

Rise/Fall Delay with External Load

$$t_r = k_1 \frac{R_{pu}}{s}\left(sC_{int} + C_{ext}\right)$$

$$= k_1 \underbrace{R_{pu}C_{int}}_{t_{ri}}\left(1 + \frac{C_{ext}}{sC_{int}}\right)$$

$$= t_{ri}\left(1 + \frac{C_{ext}}{sC_{int}}\right)$$

$$t_f = t_{fu}\left(1 + \frac{C_{ext}}{sC_{int}}\right)$$

Impact of Sizing

Propagation delay of a unit inverter

$$t_p = 0.69\, R_{eq}\,(C_{int} + C_{ext})$$

$$= 0.69\, R_{eq} C_{int} \left(1 + \frac{C_{ext}}{C_{int}}\right) = t_{p0} \left(1 + \frac{C_{ext}}{C_{int}}\right)$$

t_{p0}: intrinsic or unloaded delay

- R_{eq} : equivalent on-resistance of the inverter
- C_{int} : intrinsic capacitance (gate, diffusion, wiring, etc.)
- C_{ext} : external load capacitance
- t_{p0} : intrinsic (unloaded) delay

Effect of sizing (S)

S: sizing factor w.r.t. a reference unit-size inverter.

$$t_p = 0.69 \left(\frac{R_{ref}}{S}\right)(SC_{int,ref}) \left(1 + \frac{C_{ext}}{SC_{int,ref}}\right)$$

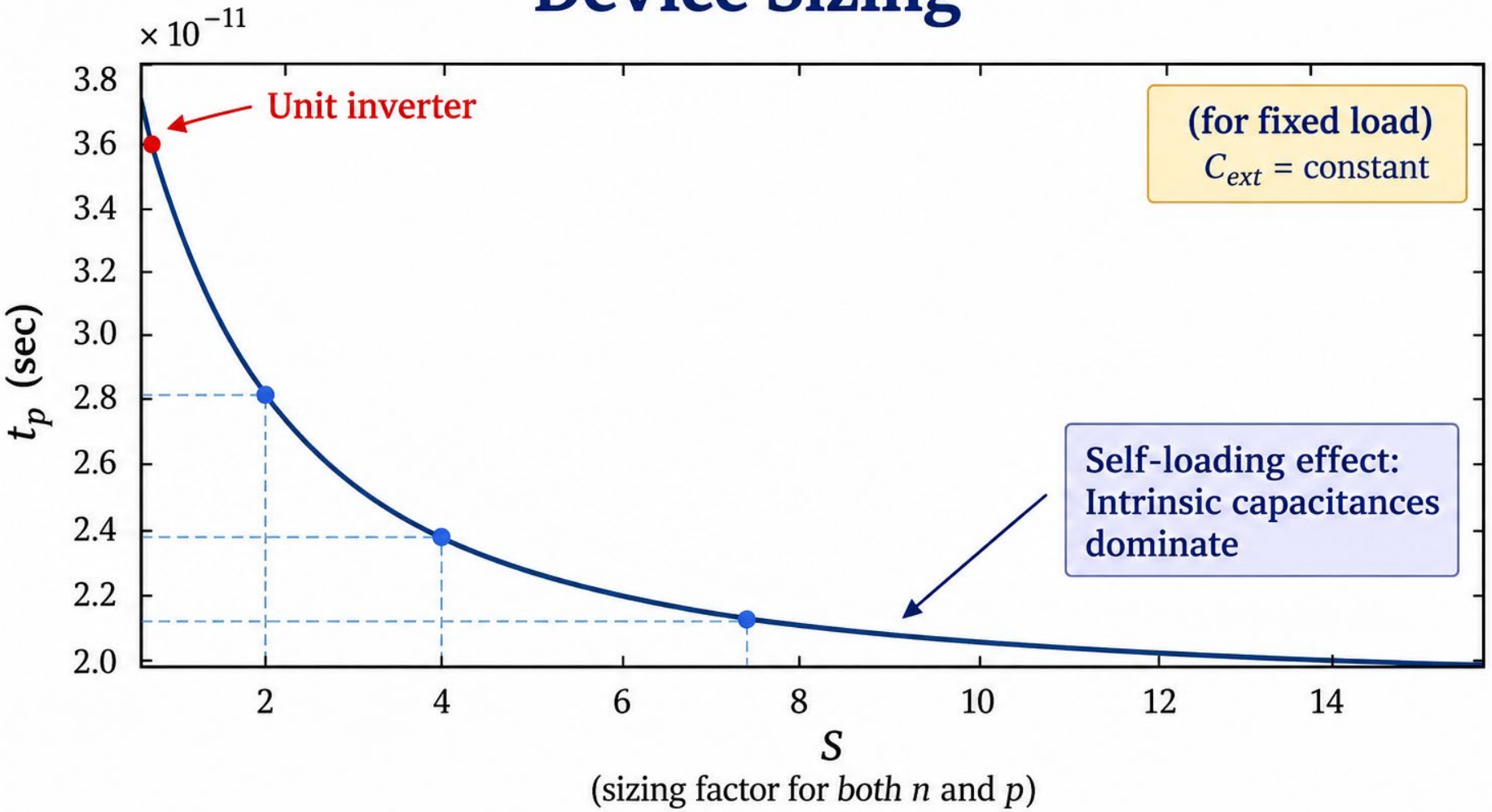
$$= 0.69\, R_{ref} C_{int,ref} \left(1 + \frac{C_{ext}}{SC_{int,ref}}\right) = t_{p0} \left(1 + \frac{C_{ext}}{SC_{int,ref}}\right)$$

Increasing the sizing factor S reduces the impact of the external load by a factor of 1/S.

Key Takeaways

- **Intrinsic delay is independent of gate sizing.**
  The intrinsic delay t_{p0} is determined by technology and inverter layout. When no load is present, an increase in the drive (larger S) is totally offset by the increased capacitance.
- **Very large sizing yields diminishing returns.**
  Making S infinitely large yields the maximum obtaainable performance gain, eliminating the impact of any external load, and reducing the delay to the intrinsic one. However, any sizing factor S that is sufficiently larger than (C_{ext}/C_{int}) produces similar results at a substantial gain in silicon area.

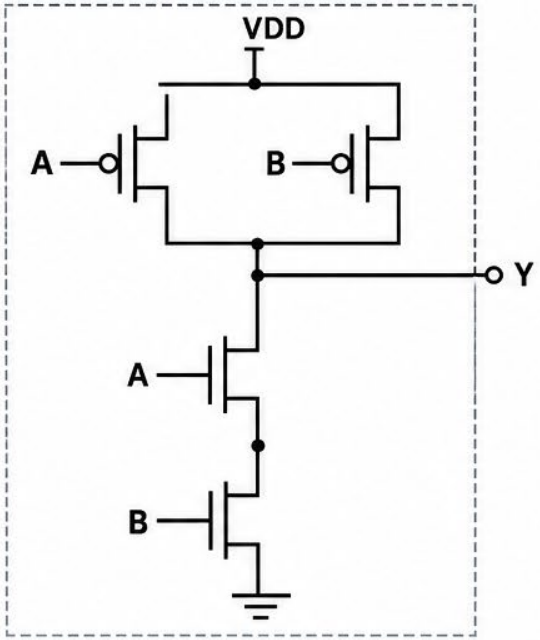
Device Sizing



Takeaway: With a fixed external load, increasing the device size S reduces the delay, but with diminishing returns as the intrinsic (self-loading) capacitances dominate.

Timing Arcs in a CMOS NAND2 Cell

CMOS NAND2 Schematic



$Y = \neg(A \cdot B)$
(NAND2)

Timing Arcs

- A → Y : t_{PLH} ($A \rightarrow Y$)
- A → Y : t_{PHL} ($A \rightarrow Y$)
- B → Y : t_{PLH} ($B \rightarrow Y$)
- B → Y : t_{PHL} ($B \rightarrow Y$)

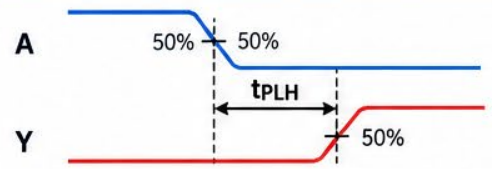
Propagation delay arcs (combinational)

- A → Y : t_{setup} (A)
- A → Y : t_{hold} (A)
- B → Y : t_{setup} (B)
- B → Y : t_{hold} (B)

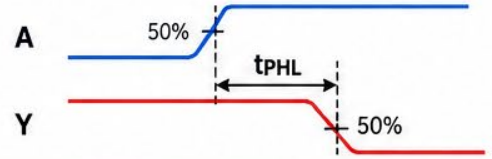
Timing check arcs (for sequential endpoints such as flip-flops)

Propagation Delay Arcs (Waveforms)

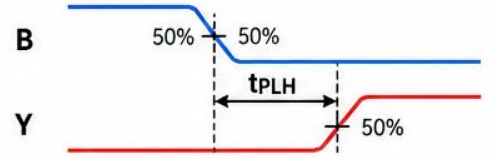
A → Y : t_{PLH} (output rising)



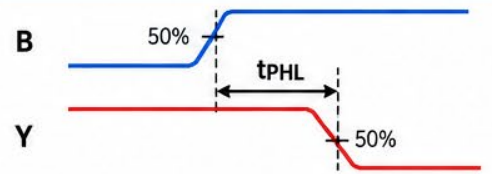
A → Y : t_{PHL} (output falling)



B → Y : t_{PLH} (output rising)



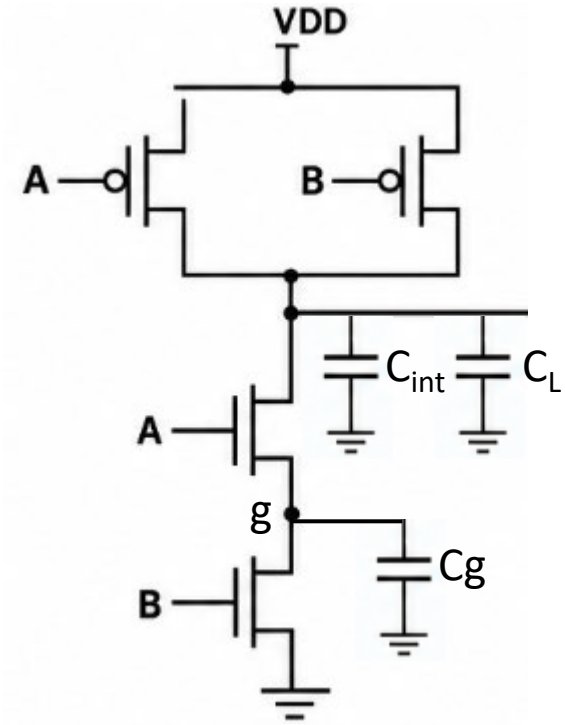
B → Y : t_{PHL} (output falling)



Key Points

- A CMOS NAND2 has four propagation arcs: two from each input (A, B) to the output (Y).
- t_{PLH}: delay from input falling (or rising, depending on arc) to output low → high transition.
- t_{PHL}: delay from input rising (or falling) to output high → low transition.
- Each input has separate arcs because delays can be different (due to transistor sizing, stack effect, etc.).
- Timing check arcs (setup/hold) are defined when the NAND2 output drives a sequential element (e.g., flip-flop).
- Delays are measured between 50% voltage points of the input and output waveforms.

Propagation Delays for Different Timing Arcs



Arc	First input vector (Setup)	Second input vector (Evaluate)	Intrinsic Delay (ps)
$t_{A\rightarrow Y\uparrow}$			
$t_{A\rightarrow Y\downarrow}$			
$t_{B\rightarrow Y\uparrow}$			
$t_{B\rightarrow Y\downarrow}$			

For each arc, propagation delays and rise/fall times depend on the output load and input slew.

Timing Data for a Cell

For each arc, propagation delays and rise/fall times depend on the output load and input slew.

t_{pLH}: $t_{A \to Y\uparrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

t_r: $t_{A \to Y\uparrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

t_{pHL}: $t_{A \to Y\downarrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

t_f: $t_{A \to Y\downarrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

t_{pLH}: $t_{B \to Y\uparrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

t_r: $t_{B \to Y\uparrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

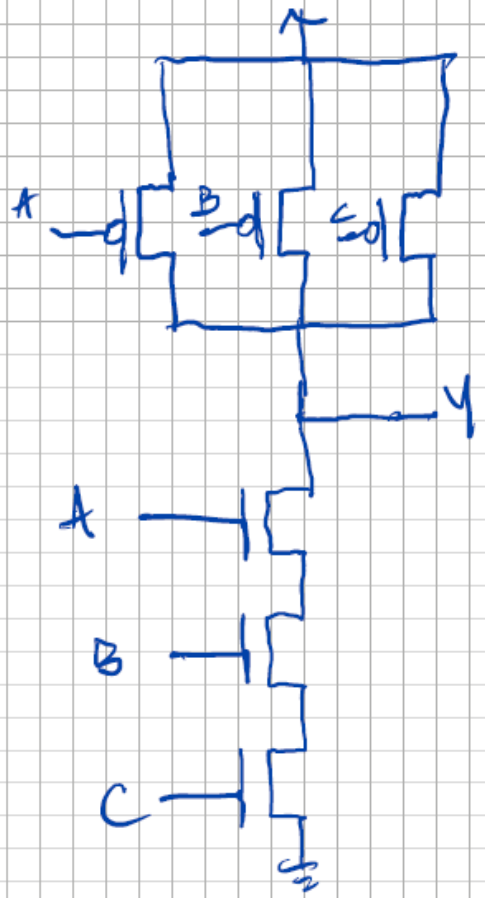
t_{pHL}: $t_{B \to Y\downarrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

t_f: $t_{B \to Y\downarrow}$ Input: <vector 1> → <vector 2>

Slew (ps) / Load (fF)	10	30	50	70
0				
3				
6				
9				

Practice.



1) Determine Timing Arcs

2) For each arc, find Set up and eval. vectors

3) For each arc, Determine all charged/discharged caps. Draw the RC Circuit.